Graphs

1. Data structures recap: For each of the following claims, explain what is missing or incorrect, or defend why the statement makes sense.

   (a) "Linked lists are best when there are many insertions, because insertion takes $O(1)$ time."

   (b) "Hash tables are always better than AVL trees, because $O(1)$ is better than $O(\log n)$."

   (c) "Arrays are poor choices for changing collections, because every insertion takes $O(n)$ time."

2. Consider the adjacency matrix and adjacency list representations of a graph. Which choice would be better in each of the following situations? Justify your answer.

   (a) The graph has 10,000 vertices and 20,000 edges, and it is important to use as little space as possible.

   (b) The graph has 10,000 vertices and 20,000,000 edges, and it is important to use as little space as possible.

   (c) You need to answer queries to find if two vertices are adjacent as fast as possible, no matter how much space you use.